

Hodgkin lymphoma

Highly curable, and managed by **interim PET** more than by anything else — the whole game is giving enough therapy to cure while withdrawing bleomycin and radiation from the people who don't need them

Stanford Oncology Handbook 2025, Lymphoma: Hodgkin, pp.181–192 (Chou L, Schroers-Martin J) · NCCN v2.2025 · items marked † are outside the handbook chapter

WORKUP

- History: B symptoms, **pruritus**, **alcohol intolerance**; exam of nodal regions, liver, spleen. **Marrow biopsy only** for focal bone lesions or unexplained cytopenias
- Excisional biopsy preferred**, core acceptable, **no FNA**
- IHC — most critical: **CD30**, **CD15**, **PAX5**, **CD20**, **CD3**
- Labbs: CBC, CMP, **ESR**, LDH, HIV, hepatitis B and C
- PET-CT** for staging
- Echo/MUGA** before anthracycline · **PFTs** before bleomycin · before a checkpoint inhibitor: TSH, ACTH, cortisol, troponin
- Fertility discussion — risk is **low with A(B)VD**, so preservation is not mandated frontline; bank before salvage

STAGING — COTSWOLDS-MODIFIED ANN ARBOR

I	One lymph node region or lymphoid structure (spleen, thymus, Waldeyer ring)
II	≥2 regions, same side of the diaphragm. Hilar nodes are lateralized — bilateral involvement is stage II
III	Regions on both sides of the diaphragm
IV	Disseminated — marrow, CNS, or more than one extranodal site
A / B	B = weight loss >10% over 6 mo, or unexplained fever >38°C in the past month, or recurrent drenching night sweats
S · X · E	S splenic · X bulky : node ≥10 cm , or mediastinal mass ≥½ the transverse thoracic diameter at T5–6 · E limited direct extranodal extension

A single extranodal site as the only disease is IE; more than one extranodal site is stage IV. Hilar (bronchopulmonary) nodes sit outside the mediastinum.

EARLY-STAGE RISK FACTORS

Factor	EORTC	GHSG	NCCN
Mediastinal mass	>0.35 CXR	>½ CXR	>½ or >10 cm
Age ≥50	yes	—	—
ESR	≥50 (≥30 if B)	≥50 (≥30 if B)	≥50 or any B
Extranodal disease	—	yes	—
Nodal areas	≥3	≥2	≥3

Favorable = CS I–II with none of these. GHSG counts nodal areas differently from Ann Arbor (infraclavicular with ipsilateral cervical, both hila with mediastinum, abdomen split in two).

IPS — ADVANCED STAGE

Factors	FFP Hasenclever	FFP Moccia
0–1	84–77%	88–84%
2–3	67–60%	80–74%
≥4	51–42%	67–62%

One point each: **albumin <4** · **Hgb <10.5** · **male** · **stage IV** · **age ≥45** · **WBC ≥15,000** · **lymphocytes <600** or <8%. Hasenclever NEJM 1998; Moccia JCO 2012 is the modern-era revision.

TREATMENT PATHWAY

Early favorable (CS I–II, no risk factors)

↓

PET-adapted, two routes
GHSG-favorable: **ABVD ×2 → 20 Gy ISRT** (HD16)
otherwise ABVD ×3 → **30 Gy** if DS 1–2
chemo only: **ABVD ×4** if DS 1–2
DS 3 → ABVD ×2 then AVD ×4 (RATHL)

Early unfavorable

↓

Combined modality or chemo alone
ABVD ×4 + 30 Gy ISRT if DS 1–2
nivo-AVD ×4 + 30 Gy (NIVAHL) · BV-AVD (BREACH)
chemo only: ABVD ×2 → AVD ×4 — **drop bleomycin** after cycle 2 if DS 1–3 (RATHL)

Advanced III–IV

↓

Two modern standards
BrECADD ×4–6 (HD21) — stop at 4 if DS 1–3
nivolumab-AVD ×6 (S1826)
selected: BV-AVD ×6 (ECHELON-1), ABVD

Relapsed / refractory

↓

Checkpoint combo is preferred first salvage
pembro-GVD · pembro-ICE · nivo-ICE · BV-nivo
non-CPI: BeGEV, BV ± benda, BV-ICE, DHAP, GVD, ICE
2–4 cycles then mobilize if transplant-bound

NLPHL

↓

A different disease
I–IIA contiguous non-bulky: **ISRT 30–36 Gy**
I–IIB or bulky: R-ABVD/R-CHOP ×4 + 30 Gy
III–IV: **R-CHOP ×6** ± ISRT, or observe
transformation to THRLBCL → treat as DLBCL

† DEAUVILLE 5-POINT SCORE

- 1 · 2

No uptake · uptake ≤ **mediastinum**
- 3

Uptake > **mediastinum but ≤ liver** — usually adequate response
- 4 · 5

Moderately > **liver** · markedly > liver or new lesions — **treatment failure**

† Not in the handbook chapter, which specifies only PET/CT at completion (plus marrow biopsy if positive at diagnosis). Added because response is reported this way in practice and in every trial cited here.

FRONTLINE REGIMENS

Regimen	Dosing	Cycle
ABVD	Doxorubicin 25 · bleomycin 10 units/m² · vinblastine 6 · dacarbazine 375 — all d1 + d15	q28d
Nivolumab-AVD	Nivolumab 240 mg · doxorubicin 25 · vinblastine 6 · dacarbazine 375 — all d1 + d15. G-CSF not required	d1 + d15
BrECADD	Brentuximab 1.8 d0 · etoposide 150 d1–3 · cyclophosphamide 1250 d1 · doxorubicin 40 d1 · vincristine 1.4 d8 · dacarbazine 250 d2–3 · dexamethasone 40 mg d1–4	q21d with G-CSF
BV-AVD	Brentuximab 1.2 mg/kg · doxorubicin 25 · vinblastine 6 · dacarbazine 375 — all d1 + d15; filgrastim d5–8 and d19–22	q28d

mg/m² unless stated. **BrECADD starts at dose level 4** and de-escalates for toxicity. Antiviral + PJP prophylaxis with every brentuximab-containing regimen.

SALVAGE REGIMENS

Regimen	Dosing	Cycle
Pembro-GVD	Pembrolizumab 200 mg d1; gemcitabine 1000 (800 if post-HSCT) + vinorelbine 15 + liposomal doxorubicin 10 — all d1 + d8	q21d ×2–4
Pembro-ICE · nivo-ICE	Checkpoint inhibitor d1 + ICE: ifosfamide 5000 + mesna 5000 CIV d2; carboplatin AUC 5 (max 800 mg) d2; etoposide 100 d1–3	q21d ×2, then collect
BV + nivolumab	Brentuximab 1.8 mg/kg (max 180 mg) d1; nivolumab 240 mg C1D8 then d1 of C2–4	q21d ×4
BV ± benda · BeGEV	Brentuximab 1.8 mg/kg d1 ± bendamustine 90 d1–2 (BV alone ×16 cycles) · BeGEV = gemcitabine 800 d1+4, bendamustine 90 d2+3, vinorelbine 20 d1, prednisolone 100 mg d1–4	q21d
GVD · ICE · DHAP	Gemcitabine/vinorelbine/liposomal doxorubicin · ifosfamide-carbo-etoposide · cisplatin + cytarabine 2000 q12h ×2 + dexamethasone	q21–28d

2–4 cycles then mobilization for transplant candidates, 4–6 otherwise. **Checkpoint monotherapy responds often but completes rarely** — combine it.

PEARLS

- Do not give G-CSF with ABVD** — bleomycin pulmonary toxicity. Do not delay or dose-reduce for non-febrile neutropenia; if unavoidable (ANC <200 or febrile neutropenia) give it day 5 or later ×3–5 days
- Bilateral hilar involvement is stage II**, not III — hilar nodes are lateralized, and they sit outside the mediastinum
- In relapse, give a **checkpoint combination first if the patient is CPI-naïve** — post-ASCT outcomes are better
- † The handbook defers survivorship to NCCN — cardiac late effects, second malignancy and **post-chest-RT breast screening** are not in it

TRIALS BEHIND THE PATHWAY

Trial	Role
HD16	ABVD ×2 + 20 Gy for GHSG-favorable early stage with a negative interim PET
RAPID · EORTC H10	PET-adapted radiation omission vs combined modality in early stage
RATHL	Drop bleomycin after 2 cycles if interim PET is negative
HD21	BrECADD vs eBEACOPP in advanced stage — the current European standard
SWOG S1826	Nivolumab-AVD vs BV-AVD in advanced stage
KEYNOTE-204	Pembrolizumab vs brentuximab, relapsed

HD16, RAPID and H10 all failed non-inferiority — omitting RT costs ~4–7% PFS, though OS is unchanged.